

This document provides additional assistance with wiring your Extron IPL PRO enabled product to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instruction, refer to the user's manual for the specific Extron IPL PRO enabled product or the controlled device manufacturer supplied documentation.

Device Specifications:

Device Type: Video Conference
Manufacturer: Cisco
Firmware Version: CE9.0
Model(s): Spark Room Kit CE9.0.X, Spark Room Kit Plus CE9.0.X

Tested on the Following Software and Firmware Versions:

IP Link Pro Control Processor Firmware	Touch Link Panel Firmware	GS Version
2.06.0000-b010	2.01.0001-b001	1.2.3

Version History

Module Version	Date	Notes
1_0_0_0	6/29/2017	Initial Version

Module Notes

- Use highest baud rate for better performance when using a serial connection.
- SpeakerTrack: The SpeakerTrack Control command will work only after the SpeakerTrack Mode: Auto command is sent. This command applies only when using a Cisco TelePresence SpeakerTrack 60 camera.
- devicePassword and deviceUsername variables must be set accordingly.
Example: InterfaceName.devicePassword = 'TANDBERG' InterfaceName.deviceUsername = 'admin'
- connectionCounter variable must be set the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15. Example:
InterfaceName.connectionCounter = 5
- CallHistoryOccurrenceType variable must be set accordingly. Default value is 'Time'. Call History Occurrence Type ranges from 'Time and 'Frequency'. Example: InterfaceName.
CallHistoryOccurrenceType = 'Time'
- NumberofPhonebookSearch variable must be set accordingly. Default value is 5. Number of Phonebook Search ranges from 1 to 10. Example: InterfaceName.NumberofPhonebookSearch = 5
- NumberofCallHistory variable must be set accordingly. Default value is 5 Number of Call History ranges from 1 to 20. Example: InterfaceName.NumberofCallHistory = 5
- PhonebookSearchOffset variable must be set accordingly. Default value is 50 PhonebookSearchOffset ranges from 20 - 250. Example: InterfaceName.PhonebookSearchOffset = 50

Phonebook commands:

- PhonebookSearchResult
This is where the search results will be saved to when phonebook is searched. The number of results are set by a driver parameter Number of Phonebook Search, this is initialized to 5 results by default. Simply call SetText with the returned result to display the search result on a button or label object.
- PhonebookSearchSet
When the number selected in this command is triggered, it will dial the number with the xCommand Dial Number command.
- PhonebookNavigation
This command is used for navigating through the search results. The navigation commands scroll through the results when there are more results than NumberofPhonebookSearch variable.
- PhonebookUpdate
This is used to trigger the Phonebook search.
- PhonebookFolderIDUpdate
This is used to trigger the Folder search within the Phonebook.
- PhonebookFolderIDSearchSet
When the Folder Name selected in this command is triggered, it will store the Folder Name in a class variable, FolderIDNumber, and that variable can be accessed and used in the 'FolderID' qualifier in the PhonebookUpdate command.
- PhonebookFolderIDSearchResult
This is where the search results are saved when PhonebookFolderIDUpdate is called. The number of results displayed is set to 5 by default. Simply call SetText with the returned results to display the results on a button or label object.

Call History commands:

- CallHistoryRefresh
 - This is used to trigger the search for recent Call History based on the CallHistoryOccuranceType class variable.
- CallHistoryNavigation
 - This command is used for navigating through the CallHistory results.
- CallHistorySelect
 - When the CallHistorySelect is triggered, it will store the call history results' Callback Number into a class variable name, number. It can be accessed and use in the 'Number' qualifier for the Hook Dial command.
- CallHistory
 - This is where the search results are saved when CallHistoryRefresh is called. The number of results displayed is set to 5 by default. Simply call SetText to display the results on a button or label object.

Control Commands, States, and Qualifiers:

AudioOutput	'On'	'Off'	
['Output']	'Internal Speaker'	'Line'	
AutoAnswer	'On'	'Off'	
CallHistoryNavigation	'Up'	'Down'	'Page Up'
	'Page Down'		
CallHistoryRefresh	None		
CallHistorySelect	'1' – '20'		
CallSetupMode	'Gatekeeper'	'Direct'	
CameraFocus	'Near'	'Far'	'Stop'
	'Auto'		
CameraPan	'Left'	'Right'	'Stop'
['Speed']	1 to 15 in steps of 1		
CameraPresetRecall	'1' – '35'		
CameraPresetSave	'1' – '35'		
CameraTilt	'Up'	'Down'	'Stop'
['Speed']	1 to 15 in steps of 1		
CameraZoom	'In'	'Out'	'Stop'
['Speed']	1 to 15 in steps of 1		
ConnectedDeviceNavigation	'Up'	'Down'	'Page Up'
	'Page Down'		
ConnectedDeviceUpdate	None		
DisplayMode	'Dual'	'Single'	'Dual Presentation Only'
	'Auto'		
DNSDomainNameCommand	'String'		
DNSServerAddressCommand	'String'		
DoNotDisturb	'Active'	'Inactive'	
DTMF	'0'	'1'	'2'
	'3'	'4'	'5'
	'6'	'7'	'8'
	'9'	'*'	'#'
FarEndCameraPanTilt	'Up'	'Down'	'Left'
	'Right'	'Stop'	

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FarEndCameraPresetRecall	'1' – '15'		
FarEndCameraSource	'0' – '15'		
FarEndCameraZoom	'In'	'Out'	'Stop'
FarEndControl	'On'	'Off'	
H323GatekeeperAddressCommand	'String'		
H323AliasE164Command	'String'		
H323AliasIDCommand	'String'		
H323ProfileAuthenticationLoginNameCommand	'String'		
H323ProfileAuthenticationPasswordCommand	'String'		
Hook	'Dial'	'Accept'	'Join 1'
	'Join 2'	'Join 3'	'Join 4'
	'Join 5'	'Hold 1'	'Hold 2'
	'Hold 3'	'Hold 4'	'Hold 5'
	'Resume 1'	'Resume 2'	'Resume 3'
	'Resume 4'	'Resume 5'	'Disconnect 1'
	'Disconnect 2'	'Disconnect 3'	'Disconnect 4'
	'Disconnect 5'	'Reject'	
['Number']	'String'		
['Protocol']	'Auto'	'SIP'	'H323'
	'H320'	'Spark'	
Input	'1' – '5'		
InputMicLevel	0 to 24 in steps of 1		
['Input']	'Microphone 1'	'Microphone 2'	
InputMute	'On'	'Off'	
['Input']	'Internal Microphone'	'Microphone 1'	'Microphone 2'
IPv4AddressCommand	'String'		
IPv4GatewayCommand	'String'		
IPv4SubnetMaskCommand	'String'		
IREmulation	'0'	'1'	'2'
	'3'	'4'	'5'
	'6'	'7'	'8'

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	'9'	'C'	'Call'
	'Down'	'F1'	'F2'
	'F3'	'F4'	'F5'
	'Grab'	'Home'	'Layout'
	'Left'	'Mute'	'Mute Mic'
	'Ok'	'PhoneBook'	'Presentation'
	'Right'	'Selfview'	'Square'
	'Aux'	'Camera'	'Doc Cam'
	'PC'	'VCR'	'*'
	'Up'	'Volume Down'	'Volume Up'
	'Zoom In'	'Zoom Out'	'Disconnect'
	'+'	'_'	
['Key']	'Click'	'Press'	'Release'
LayoutSet	'Auto'	'Custom'	'Equal'
	'Overlay'	'Prominent'	'Single'
['Target']	'Local'	'Remote'	
MicrophonesMute	'On'	'Off'	
PhonebookFolderIDNavigation	'Up'	'Down'	'Page Up'
	'Page Down'		
PhonebookFolderIDSearchSet	1 – 10		
PhonebookFolderIDUpdate	None		
['Phonebook Type']	'Corporate'	'Local'	
PhonebookNavigation	'Up'	'Down'	'Page Up'
	'Page Down'	'Previous Folder'	
['Contact']	String		
['FolderID']	String		
['Phonebook Type']	'Corporate'	'Local'	
PhonebookSearchSet	1 – 10		
PhonebookUpdate	'Refresh'	'Next Search'	'Previous Search'
['Contact']	String		
['FolderID']	String		
['Phonebook Type']	'Corporate'	'Local'	
Presentation	'Start'	'Stop'	
['Instance']	'1' – '6'		

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PresentationSendingMode	'Local and Remote'	'Local Only'	
PresetRecall	'1' – '15'		
PresetSave	'1' – '15'		
Reboot	None		
SelfView	'On'	'Off'	
SelfViewDefaultFullscreenMode	'On'	'Off'	'Current'
SelfViewPosition	'Upper Left' 'Center Left' 'Lower Right'	'Upper Center' 'Center Right'	'Upper Right' 'Lower Left'
SleepTimer	1 – 480		
SpeakerTrackControl	'On'	'Off'	
SpeakerTrackMode	'Auto'	'Off'	
Standby	'Activate'	'Deactivate'	
Volume	0 to 100 in steps of 1		

Status Available:

AudioOutput	'On'	'Off'	
['Output']	'Internal Speaker'	'Line'	
AutoAnswer	'On'	'Off'	
CallHistory	'String'		
['Button']	'1' – '20'		
['Detail Type']	'Display Name'	'Callback Number'	'Last Occurrence Time'
	'Occurrence Count'	'Occurrence Type'	
CallSetupMode	'Gatekeeper'	'Direct'	
CallStatus	'Dialing'	'Connecting'	'Connected'
	'Ringing'	'Idle'	'Disconnecting'
	'On Hold'	'Early Media'	'Preserved'
	'Remote Preserved'		
['Call']	'1' – '5'		
CallStatusType	'Video'	'Audio'	'Audio Can Escalate'
	'Forward All Call'	'Unknown'	
['Call']	'1' – '5'		
ConnectedDeviceMAC	'String'		
['Button']	1 to 10 in steps of 1		
ConnectedDeviceName	'String'		
['Button']	1 to 10 in steps of 1		
ConnectedDeviceType	'String'		
['Button']	1 to 10 in steps of 1		
ConnectionStatus	'Connected'	'Disconnected'	
DisplayMode	'Dual'	'Single'	'Dual Presentation Only'
	'Triple Presentation Only'	'Triple'	'Quadruple'
DisplayName	'String'		
['Call']	'1' – '5'		
DNSDomainName	'String'		
DNSServerAddress	'String'		
['Server']	'1' – '5'		
DoNotDisturb	'Active'	'Inactive'	
FarEndControl	'On'	'Off'	
FirmwareVersion	'String'		
GatewayAddress	'String'		

H323GatekeeperAddress	'String'		
H323GatekeeperStatus	'Required'	'Discovering'	'Discovered'
	'Authenticating'	'Authenticated'	'Registering'
	'Registered'	'Inactive'	'Rejected'
Input	'1' – '5'		
InputMicLevel	0 to 24 in steps of 1		
['Input']	'Microphone 1'	'Microphone 2'	
InputMute	'On'	'Off'	
['Input']	'Internal Microphone'	'Microphone 1'	'Microphone 2'
InputSignal	'True'	'False'	'Unknown'
IPAddress	'String'		
MACAddress	'String'		
MicrophonesMute	'On'	'Off'	
NetworkAssignment	'Static'	'DHCP'	
PhonebookFolderIDSearchResult	'String'		
['Button']	1 to 10 in steps of 1		
PhonebookSearchResult	'String'		
['Button']	1 to 10 in steps of 1		
Presentation	'Start'	'Stop'	
['Instance']	'1' – '6'		
PresentationMode	'On'	'Off'	
PresentationSendingModeStatus	'Local and Remote'	'Local Only'	'Off'
['Instance']	'1' – '6'		
RemoteNumber	'String'		
['Call']	'1' – '5'		
SelfView	'On'	'Off'	
SelfViewDefaultFullscreenMode	'On'	'Off'	'Current'
SelfViewPosition	'Upper Left'	'Upper Center'	'Upper Right'
	'Center Left'	'Center Right'	'Lower Left'
	'Lower Right'		
SIPRegistrationStatus	'Deregistering'	'Failed'	'Inactive'
	'Registered'	'Registering'	
SpeakerTrackControl	'On'	'Off'	

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SpeakerTrackMode	'Auto'	'Off'
Standby	'Activate'	'Deactivate'
SubnetMask	'String'	
Volume	0 to 100 in steps of 1	

Cable and Adapter Requirements:

Captive Screw to USB Type-A RS-232 Serial Cable adapter

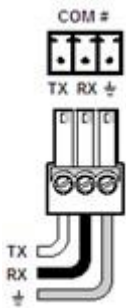
Notes for the Device:

Serial communication:

Port Type: RS-232
Baud Rate: 115200
Data Bits: 8

Parity: None
Stop Bits: One
Flow Control: None

Pin Assignments Diagram



Please contact the manufacturer for pin assignment diagram.

General Notes:

Network communication:

When configuring the Ethernet module, be sure device settings match that of the GS interface settings**gs**.

Port Type:	Ethernet
Default Port:	23
Logon Credentials Supported:	Yes
Default Username:	admin
Default Password:	TANDBERG
Multi-Connection Capabilities:	Yes
Port Changeability:	No

Ethernet Module Configuration Description:

Please refer to user manual for settings and changes to the network communication

Notes for the Device:

Telnet Mode must be set to On by an admin in order for the driver to work.

Appendix A. Set Commands

Audio Output Off Output Internal Speaker	xConfiguration Audio Output InternalSpeaker Mode:Off\x0D
Audio Output Off Output Line	xConfiguration Audio Output Line 1 Mode:Off\x0D
Audio Output On Output Internal Speaker	xConfiguration Audio Output InternalSpeaker Mode:On\x0D
Audio Output On Output Line	xConfiguration Audio Output Line 1 Mode:On\x0D
Auto Answer Off	xConfiguration Conference AutoAnswer Mode: Off\x0D
Auto Answer On	xConfiguration Conference AutoAnswer Mode: On\x0D
Call Setup Mode Direct	xConfiguration H323 CallSetup Mode: Direct\x0D
Call Setup Mode Gatekeeper	xConfiguration H323 CallSetup Mode: Gatekeeper\x0D
Camera Focus Auto	xCommand Camera TriggerAutoFocus CameraId:1\x0D
Camera Focus Far	xCommand Camera Ramp CameraId:1 Focus:Far\x0D
Camera Focus Near	xCommand Camera Ramp CameraId:1 Focus:Near\x0D
Camera Focus Stop	xCommand Camera Ramp CameraId:1 Focus:Stop\x0D
Camera Pan Left Speed 1	xCommand Camera Ramp CameraId:1 Pan:Left PanSpeed:1\x0D
Camera Pan Left Speed 15	xCommand Camera Ramp CameraId:1 Pan:Left PanSpeed:15\x0D
Camera Pan Right Speed 1	xCommand Camera Ramp CameraId:1 Pan:Right PanSpeed:1\x0D
Camera Pan Right Speed 15	xCommand Camera Ramp CameraId:1 Pan:Right PanSpeed:15\x0D
Camera Pan Stop Speed 1	xCommand Camera Ramp CameraId:1 Pan: Stop\x0D
Camera Pan Stop Speed 15	xCommand Camera Ramp CameraId:1 Pan: Stop\x0D
Camera Preset Recall 1	xCommand Camera Preset Activate PresetId: 1\x0D
Camera Preset Recall 35	xCommand Camera Preset Activate PresetId: 35\x0D
Camera Preset Save 1	xCommand Camera Preset Store PresetId:1 CameraId:1\x0D
Camera Preset Save 35	xCommand Camera Preset Store PresetId:35 CameraId:1\x0D
Camera Tilt Down Speed 1	xCommand Camera Ramp CameraId:1 Tilt:Down TiltSpeed:1\x0D
Camera Tilt Down Speed 15	xCommand Camera Ramp CameraId:1 Tilt:Down TiltSpeed:15\x0D
Camera Tilt Stop Speed 1	xCommand Camera Ramp CameraId:1 Tilt: Stop\x0D
Camera Tilt Stop Speed 15	xCommand Camera Ramp CameraId:1 Tilt: Stop\x0D
Camera Tilt Up Speed 1	xCommand Camera Ramp CameraId:1 Tilt:Up TiltSpeed:1\x0D
Camera Tilt Up Speed 15	xCommand Camera Ramp CameraId:1 Tilt:Up TiltSpeed:15\x0D
Camera Zoom In Speed 1	xCommand Camera Ramp CameraId:1 Zoom:In ZoomSpeed:1\x0D
Camera Zoom In Speed 15	xCommand Camera Ramp CameraId:1 Zoom:In ZoomSpeed:15\x0D
Camera Zoom Out Speed 1	xCommand Camera Ramp CameraId:1 Zoom:Out ZoomSpeed:1\x0D
Camera Zoom Out Speed 15	xCommand Camera Ramp CameraId:1 Zoom:Out ZoomSpeed:15\x0D
Camera Zoom Stop Speed 1	xCommand Camera Ramp CameraId:1 Zoom: Stop\x0D
Camera Zoom Stop Speed 15	xCommand Camera Ramp CameraId:1 Zoom: Stop\x0D
Display Mode Auto	xConfiguration Video Monitors: Auto\x0D
Display Mode Dual	xConfiguration Video Monitors: Dual\x0D
Display Mode Dual Presentation Only	xConfiguration Video Monitors: DualPresentationOnly\x0D
Display Mode Single	xConfiguration Video Monitors: Single\x0D
Do Not Disturb Active	xCommand Conference DoNotDisturb Activate\x0D
Do Not Disturb Inactive	xCommand Conference DoNotDisturb Deactivate\x0D
DTMF #	xCommand Call DTMFSend DTMFString:#\x0D
DTMF *	xCommand Call DTMFSend DTMFString:*\x0D
DTMF 0	xCommand Call DTMFSend DTMFString:0\x0D
DTMF 9	xCommand Call DTMFSend DTMFString:9\x0D

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Far End Camera Pan/Tilt Down	xCommand Call FarEndControl Camera Move Value:Down\x0D
Far End Camera Pan/Tilt Left	xCommand Call FarEndControl Camera Move Value:Left\x0D
Far End Camera Pan/Tilt Right	xCommand Call FarEndControl Camera Move Value:Right\x0D
Far End Camera Pan/Tilt Stop	xCommand Call FarEndControl Camera Stop\x0D
Far End Camera Pan/Tilt Up	xCommand Call FarEndControl Camera Move Value:Up\x0D
Far End Camera Preset Recall 1	xCommand Call FarEndControl RoomPreset Activate:1\x0D
Far End Camera Preset Recall 15	xCommand Call FarEndControl RoomPreset Activate:15\x0D
Far End Camera Source 0	xCommand Call FarEndControl Source Select SourceId:0\x0D
Far End Camera Source 15	xCommand Call FarEndControl Source Select SourceId:15\x0D
Far End Camera Zoom In	xCommand Call FarEndControl Camera Move Value:ZoomIn\x0D
Far End Camera Zoom Out	xCommand Call FarEndControl Camera Move Value:ZoomOut\x0D
Far End Camera Zoom Stop	xCommand Call FarEndControl Camera Stop\x0D
Far End Control Off	xConfiguration Conference FarEndControl Mode: Off\x0D
Far End Control On	xConfiguration Conference FarEndControl Mode: On\x0D
Hook Accept Protocol Auto	xCommand Call Accept\x0D
Hook Accept Protocol H320	xCommand Call Accept\x0D
Hook Accept Protocol H323	xCommand Call Accept\x0D
Hook Accept Protocol SIP	xCommand Call Accept\x0D
Hook Accept Protocol Spark	xCommand Call Accept\x0D
Hook Reject Protocol Auto	xCommand Call Reject\x0D
Hook Reject Protocol H320	xCommand Call Reject\x0D
Hook Reject Protocol H323	xCommand Call Reject\x0D
Hook Reject Protocol SIP	xCommand Call Reject\x0D
Hook Reject Protocol Spark	xCommand Call Reject\x0D
Input 1	xCommand Video Input SetMainVideoSource ConnectorId: 1\x0D
Input 5	xCommand Video Input SetMainVideoSource ConnectorId: 5\x0D
Input Mic Level 0 Input Microphone 1	xConfiguration Audio Input 2 Level:0\x0D
Input Mic Level 0 Input Microphone 2	xConfiguration Audio Input 3 Level:0\x0D
Input Mic Level 24 Input Microphone 1	xConfiguration Audio Input 2 Level:24\x0D
Input Mic Level 24 Input Microphone 2	xConfiguration Audio Input 3 Level:24\x0D
Input Mute Off Input Internal Microphone	xConfiguration Audio Input 1 Mode:Off\x0D
Input Mute Off Input Microphone 1	xConfiguration Audio Input 2 Mode:Off\x0D
Input Mute Off Input Microphone 2	xConfiguration Audio Input 3 Mode:Off\x0D
Input Mute On Input Internal Microphone	xConfiguration Audio Input 1 Mode:On\x0D
Input Mute On Input Microphone 1	xConfiguration Audio Input 2 Mode:On\x0D
Input Mute On Input Microphone 2	xConfiguration Audio Input 3 Mode:On\x0D
IR Emulation - Key Click	xCommand UserInterface OSD Key Click Key: -\x0D
IR Emulation - Key Press	xCommand UserInterface OSD Key Press Key: -\x0D
IR Emulation - Key Release	xCommand UserInterface OSD Key Release Key: -\x0D
IR Emulation * Key Click	xCommand UserInterface OSD Key Click Key: Star\x0D
IR Emulation * Key Press	xCommand UserInterface OSD Key Press Key: Star\x0D
IR Emulation * Key Release	xCommand UserInterface OSD Key Release Key: Star\x0D
IR Emulation + Key Click	xCommand UserInterface OSD Key Click Key: +\x0D

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IR Emulation + Key Press	xCommand UserInterface OSD Key Press Key: +\x0D
IR Emulation + Key Release	xCommand UserInterface OSD Key Release Key: +\x0D
IR Emulation 0 Key Click	xCommand UserInterface OSD Key Click Key: 0\x0D
IR Emulation 0 Key Press	xCommand UserInterface OSD Key Press Key: 0\x0D
IR Emulation 0 Key Release	xCommand UserInterface OSD Key Release Key: 0\x0D
IR Emulation 9 Key Click	xCommand UserInterface OSD Key Click Key: 9\x0D
IR Emulation 9 Key Press	xCommand UserInterface OSD Key Press Key: 9\x0D
IR Emulation 9 Key Release	xCommand UserInterface OSD Key Release Key: 9\x0D
IR Emulation Aux Key Click	xCommand UserInterface OSD Key Click Key: SrcAux\x0D
IR Emulation Aux Key Press	xCommand UserInterface OSD Key Press Key: SrcAux\x0D
IR Emulation Aux Key Release	xCommand UserInterface OSD Key Release Key: SrcAux\x0D
IR Emulation C Key Click	xCommand UserInterface OSD Key Click Key: C\x0D
IR Emulation C Key Press	xCommand UserInterface OSD Key Press Key: C\x0D
IR Emulation C Key Release	xCommand UserInterface OSD Key Release Key: C\x0D
IR Emulation Call Key Click	xCommand UserInterface OSD Key Click Key: Call\x0D
IR Emulation Call Key Press	xCommand UserInterface OSD Key Press Key: Call\x0D
IR Emulation Call Key Release	xCommand UserInterface OSD Key Release Key: Call\x0D
IR Emulation Camera Key Click	xCommand UserInterface OSD Key Click Key: SrcCamera\x0D
IR Emulation Camera Key Press	xCommand UserInterface OSD Key Press Key: SrcCamera\x0D
IR Emulation Camera Key Release	xCommand UserInterface OSD Key Release Key: SrcCamera\x0D
IR Emulation Disconnect Key Click	xCommand UserInterface OSD Key Click Key: Disconnect\x0D
IR Emulation Disconnect Key Press	xCommand UserInterface OSD Key Press Key: Disconnect\x0D
IR Emulation Disconnect Key Release	xCommand UserInterface OSD Key Release Key: Disconnect\x0D
IR Emulation Doc Cam Key Click	xCommand UserInterface OSD Key Click Key: SrcDocCam\x0D
IR Emulation Doc Cam Key Press	xCommand UserInterface OSD Key Press Key: SrcDocCam\x0D
IR Emulation Doc Cam Key Release	xCommand UserInterface OSD Key Release Key: SrcDocCam\x0D
IR Emulation Down Key Click	xCommand UserInterface OSD Key Click Key: Down\x0D
IR Emulation Down Key Press	xCommand UserInterface OSD Key Press Key: Down\x0D
IR Emulation Down Key Release	xCommand UserInterface OSD Key Release Key: Down\x0D
IR Emulation F1 Key Click	xCommand UserInterface OSD Key Click Key: F1\x0D
IR Emulation F1 Key Press	xCommand UserInterface OSD Key Press Key: F1\x0D
IR Emulation F1 Key Release	xCommand UserInterface OSD Key Release Key: F1\x0D
IR Emulation F2 Key Click	xCommand UserInterface OSD Key Click Key: F2\x0D
IR Emulation F2 Key Press	xCommand UserInterface OSD Key Press Key: F2\x0D
IR Emulation F2 Key Release	xCommand UserInterface OSD Key Release Key: F2\x0D
IR Emulation F3 Key Click	xCommand UserInterface OSD Key Click Key: F3\x0D
IR Emulation F3 Key Press	xCommand UserInterface OSD Key Press Key: F3\x0D
IR Emulation F3 Key Release	xCommand UserInterface OSD Key Release Key: F3\x0D
IR Emulation F4 Key Click	xCommand UserInterface OSD Key Click Key: F4\x0D
IR Emulation F4 Key Press	xCommand UserInterface OSD Key Press Key: F4\x0D
IR Emulation F4 Key Release	xCommand UserInterface OSD Key Release Key: F4\x0D
IR Emulation F5 Key Click	xCommand UserInterface OSD Key Click Key: F5\x0D
IR Emulation F5 Key Press	xCommand UserInterface OSD Key Press Key: F5\x0D
IR Emulation F5 Key Release	xCommand UserInterface OSD Key Release Key: F5\x0D
IR Emulation Grab Key Click	xCommand UserInterface OSD Key Click Key: Grab\x0D
IR Emulation Grab Key Press	xCommand UserInterface OSD Key Press Key: Grab\x0D
IR Emulation Grab Key Release	xCommand UserInterface OSD Key Release Key: Grab\x0D

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IR Emulation Home Key Click	xCommand UserInterface OSD Key Click Key: Home\x0D
IR Emulation Home Key Press	xCommand UserInterface OSD Key Press Key: Home\x0D
IR Emulation Home Key Release	xCommand UserInterface OSD Key Release Key: Home\x0D
IR Emulation Layout Key Click	xCommand UserInterface OSD Key Click Key: Layout\x0D
IR Emulation Layout Key Press	xCommand UserInterface OSD Key Press Key: Layout\x0D
IR Emulation Layout Key Release	xCommand UserInterface OSD Key Release Key: Layout\x0D
IR Emulation Left Key Click	xCommand UserInterface OSD Key Click Key: Left\x0D
IR Emulation Left Key Press	xCommand UserInterface OSD Key Press Key: Left\x0D
IR Emulation Left Key Release	xCommand UserInterface OSD Key Release Key: Left\x0D
IR Emulation Mute Key Click	xCommand UserInterface OSD Key Click Key: Mute\x0D
IR Emulation Mute Key Press	xCommand UserInterface OSD Key Press Key: Mute\x0D
IR Emulation Mute Key Release	xCommand UserInterface OSD Key Release Key: Mute\x0D
IR Emulation Mute Mic Key Click	xCommand UserInterface OSD Key Click Key: Mute Mic\x0D
IR Emulation Mute Mic Key Press	xCommand UserInterface OSD Key Press Key: Mute Mic\x0D
IR Emulation Mute Mic Key Release	xCommand UserInterface OSD Key Release Key: Mute Mic\x0D
IR Emulation Ok Key Click	xCommand UserInterface OSD Key Click Key: Ok\x0D
IR Emulation Ok Key Press	xCommand UserInterface OSD Key Press Key: Ok\x0D
IR Emulation Ok Key Release	xCommand UserInterface OSD Key Release Key: Ok\x0D
IR Emulation PC Key Click	xCommand UserInterface OSD Key Click Key: SrcPc\x0D
IR Emulation PC Key Press	xCommand UserInterface OSD Key Press Key: SrcPc\x0D
IR Emulation PC Key Release	xCommand UserInterface OSD Key Release Key: SrcPc\x0D
IR Emulation PhoneBook Key Click	xCommand UserInterface OSD Key Click Key: PhoneBook\x0D
IR Emulation PhoneBook Key Press	xCommand UserInterface OSD Key Press Key: PhoneBook\x0D
IR Emulation PhoneBook Key Release	xCommand UserInterface OSD Key Release Key: PhoneBook\x0D
IR Emulation Presentation Key Click	xCommand UserInterface OSD Key Click Key: Presentation\x0D
IR Emulation Presentation Key Press	xCommand UserInterface OSD Key Press Key: Presentation\x0D
IR Emulation Presentation Key Release	xCommand UserInterface OSD Key Release Key: Presentation\x0D
IR Emulation Right Key Click	xCommand UserInterface OSD Key Click Key: Right\x0D
IR Emulation Right Key Press	xCommand UserInterface OSD Key Press Key: Right\x0D
IR Emulation Right Key Release	xCommand UserInterface OSD Key Release Key: Right\x0D
IR Emulation Selfview Key Click	xCommand UserInterface OSD Key Click Key: Selfview\x0D
IR Emulation Selfview Key Press	xCommand UserInterface OSD Key Press Key: Selfview\x0D
IR Emulation Selfview Key Release	xCommand UserInterface OSD Key Release Key: Selfview\x0D
IR Emulation Square Key Click	xCommand UserInterface OSD Key Click Key: Square\x0D
IR Emulation Square Key Press	xCommand UserInterface OSD Key Press Key: Square\x0D
IR Emulation Square Key Release	xCommand UserInterface OSD Key Release Key: Square\x0D
IR Emulation Up Key Click	xCommand UserInterface OSD Key Click Key: Up\x0D
IR Emulation Up Key Press	xCommand UserInterface OSD Key Press Key: Up\x0D
IR Emulation Up Key Release	xCommand UserInterface OSD Key Release Key: Up\x0D
IR Emulation VCR Key Click	xCommand UserInterface OSD Key Click Key: SrcVcr\x0D
IR Emulation VCR Key Press	xCommand UserInterface OSD Key Press Key: SrcVcr\x0D
IR Emulation VCR Key Release	xCommand UserInterface OSD Key Release Key: SrcVcr\x0D
IR Emulation Volume Down Key Click	xCommand UserInterface OSD Key Click Key: VolumeDown\x0D
IR Emulation Volume Down Key Press	xCommand UserInterface OSD Key Press Key: VolumeDown\x0D

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IR Emulation Volume Down Key Release	xCommand UserInterface OSD Key Release Key: VolumeDown\x0D
IR Emulation Volume Up Key Click	xCommand UserInterface OSD Key Click Key: VolumeUp\x0D
IR Emulation Volume Up Key Press	xCommand UserInterface OSD Key Press Key: VolumeUp\x0D
IR Emulation Volume Up Key Release	xCommand UserInterface OSD Key Release Key: VolumeUp\x0D
IR Emulation Zoom In Key Click	xCommand UserInterface OSD Key Click Key: ZoomIn\x0D
IR Emulation Zoom In Key Press	xCommand UserInterface OSD Key Press Key: ZoomIn\x0D
IR Emulation Zoom In Key Release	xCommand UserInterface OSD Key Release Key: ZoomIn\x0D
IR Emulation Zoom Out Key Click	xCommand UserInterface OSD Key Click Key: ZoomOut\x0D
IR Emulation Zoom Out Key Press	xCommand UserInterface OSD Key Press Key: ZoomOut\x0D
IR Emulation Zoom Out Key Release	xCommand UserInterface OSD Key Release Key: ZoomOut\x0D
Layout Set Auto Target Local	xCommand Video Layout LayoutFamily Set Target: local LayoutFamily:auto\x0D
Layout Set Auto Target Remote	xCommand Video Layout LayoutFamily Set Target: remote LayoutFamily:auto\x0D
Layout Set Custom Target Local	xCommand Video Layout LayoutFamily Set Target: local LayoutFamily:custom\x0D
Layout Set Custom Target Remote	xCommand Video Layout LayoutFamily Set Target: remote LayoutFamily:custom\x0D
Layout Set Equal Target Local	xCommand Video Layout LayoutFamily Set Target: local LayoutFamily:equal\x0D
Layout Set Equal Target Remote	xCommand Video Layout LayoutFamily Set Target: remote LayoutFamily:equal\x0D
Layout Set Overlay Target Local	xCommand Video Layout LayoutFamily Set Target: local LayoutFamily:overlay\x0D
Layout Set Overlay Target Remote	xCommand Video Layout LayoutFamily Set Target: remote LayoutFamily:overlay\x0D
Layout Set Prominent Target Local	xCommand Video Layout LayoutFamily Set Target: local LayoutFamily:prominent\x0D
Layout Set Prominent Target Remote	xCommand Video Layout LayoutFamily Set Target: remote LayoutFamily:prominent\x0D
Layout Set Single Target Local	xCommand Video Layout LayoutFamily Set Target: local LayoutFamily:single\x0D
Layout Set Single Target Remote	xCommand Video Layout LayoutFamily Set Target: remote LayoutFamily:single\x0D
Microphones Mute Off	xCommand Audio Microphones Unmute\x0D
Microphones Mute On	xCommand Audio Microphones Mute\x0D
Presentation Sending Mode Local and Remote	xCommand Presentation Start SendingMode: LocalRemote\x0D
Presentation Sending Mode Local Only	xCommand Presentation Start SendingMode: LocalOnly\x0D
Presentation Start Instance 1	xCommand Presentation Start Instance:1\x0D
Presentation Start Instance 6	xCommand Presentation Start Instance:6\x0D
Presentation Stop Instance 1	xCommand Presentation Stop Instance:1\x0D
Presentation Stop Instance 6	xCommand Presentation Stop Instance:6\x0D
Preset Recall 1	xCommand RoomPreset Activate PresetId:1\x0D
Preset Recall 15	xCommand RoomPreset Activate PresetId:15\x0D

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Preset Save 1	xCommand RoomPreset Store PresetId:1 Type:All\x0D
Preset Save 15	xCommand RoomPreset Store PresetId:15 Type:All\x0D
Reboot None	xCommand SystemUnit Boot Action:Restart\x0D
Self View Default Fullscreen Mode Current	xConfiguration Video Selfview Default FullscreenMode:Current\x0D
Self View Default Fullscreen Mode Off	xConfiguration Video Selfview Default FullscreenMode:Off\x0D
Self View Default Fullscreen Mode On	xConfiguration Video Selfview Default FullscreenMode:On\x0D
Self View Off	xCommand Video Selfview Set Mode:Off\x0D
Self View On	xCommand Video Selfview Set Mode:On\x0D
Self View Position Center Left	xCommand Video Selfview Set PIPPosition:CenterLeft\x0D
Self View Position Center Right	xCommand Video Selfview Set PIPPosition:CenterRight\x0D
Self View Position Lower Left	xCommand Video Selfview Set PIPPosition:LowerLeft\x0D
Self View Position Lower Right	xCommand Video Selfview Set PIPPosition:LowerRight\x0D
Self View Position Upper Center	xCommand Video Selfview Set PIPPosition:UpperCenter\x0D
Self View Position Upper Left	xCommand Video Selfview Set PIPPosition:UpperLeft\x0D
Self View Position Upper Right	xCommand Video Selfview Set PIPPosition:UpperRight\x0D
Sleep Timer 1	xCommand Standby ResetTimer Delay:1\x0D
Sleep Timer 480	xCommand Standby ResetTimer Delay:480\x0D
SpeakerTrack Control Off	xCommand Cameras SpeakerTrack Deactivate\x0D
SpeakerTrack Control On	xCommand Cameras SpeakerTrack Activate\x0D
SpeakerTrack Mode Auto	xConfiguration Cameras SpeakerTrack Mode: Auto\x0D
SpeakerTrack Mode Off	xConfiguration Cameras SpeakerTrack Mode: Off\x0D
Standby Activate	xCommand Standby Activate\x0D
Standby Deactivate	xCommand Standby Deactivate\x0D
Volume 0	xCommand Audio Volume Set Level:0\x0D
Volume 100	xCommand Audio Volume Set Level:100\x0D

Appendix B. Query Commands

Audio Output Output Internal Speaker	xConfiguration Audio Output InternalSpeaker Mode\x0D
Audio Output Output Line	xConfiguration Audio Output Line 1 Mode\x0D
Auto Answer	xConfiguration Conference AutoAnswer Mode\x0D
Call Setup Mode	xConfiguration H323 CallSetup Mode\x0D
Display Mode	xConfiguration Video Monitors\x0D
DNS Domain Name	xStatus Network 1 DNS Domain Name\x0D
DNS Server Address Server 1	xStatus Network 1 DNS Server 1 Address\x0D
DNS Server Address Server 5	xStatus Network 1 DNS Server 5 Address\x0D
Do Not Disturb	xStatus Conference DoNotDisturb\x0D
Far End Control	xConfiguration Conference FarEndControl Mode\x0D
Firmware Version	xStatus SystemUnit Software Version\x0D
Gateway Address	xStatus Network 1 IPv4 Gateway\x0D
H323 Gatekeeper Address	xStatus H323 Gatekeeper Address\x0D
H323 Gatekeeper Status	xStatus H323 Gatekeeper Status\x0D
Input	xStatus Video Input MainVideoSource\x0D
Input Mic Level Input Microphone 1	xConfiguration Audio Input 2 Level\x0D
Input Mic Level Input Microphone 2	xConfiguration Audio Input 3 Level\x0D
Input Mute Input Internal Microphone	xConfiguration Audio Input 1 Mode\x0D
Input Mute Input Microphone 1	xConfiguration Audio Input 2 Mode\x0D
Input Mute Input Microphone 2	xConfiguration Audio Input 3 Mode\x0D
Input Signal	xStatus Video Input Connector 1 Connected\x0D
IP Address	xStatus Network 1 IPv4 Address\x0D
MAC Address	xStatus Network 1 Ethernet MacAddress\x0D
Microphones Mute	xStatus Audio Microphones\x0D
Network Assignment	xConfiguration Network 1 IPv4 Assignment\x0D
Presentation Instance 1	xStatus Conference Presentation LocalInstance 1 Source\x0D
Presentation Instance 6	xStatus Conference Presentation LocalInstance 6 Source\x0D
Presentation Mode	xStatus Conference Presentation Mode\x0D
Presentation Sending Mode Status Instance 1	xStatus Conference Presentation LocalInstance 1 SendingMode\x0D
Presentation Sending Mode Status Instance 6	xStatus Conference Presentation LocalInstance 6 SendingMode\x0D
Self View	xStatus Video Selfview Mode\x0D
Self View Default Fullscreen Mode	xConfiguration Video Selfview Default FullscreenMode\x0D
Self View Position	xStatus Video Selfview PIPPosition\x0D
SIP Registration Status	xStatus SIP Registration 1 Status\x0D
SpeakerTrack Control	xStatus Cameras SpeakerTrack Status\x0D
SpeakerTrack Mode	xConfiguration Cameras SpeakerTrack Mode\x0D
Standby	xStatus Standby State\x0D
Subnet Mask	xStatus Network 1 IPv4 SubnetMask\x0D
Volume	xStatus Audio Volume\x0D